

# JERLISA WILLIAMS

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## PROFESSIONAL SUMMARY

M.S. Computer Science candidate at Full Sail University with demonstrated experience designing and evaluating applied ML/AI systems — including NLP pipelines, computer vision models, reinforcement learning agents, and production-grade AI backends. Combines rigorous model evaluation practices with software engineering discipline to ship measurable, documented results. Seeking roles in Data Science, Software Engineering, or AI Engineering where analytical depth and system-building skills drive impact.

## EDUCATION

**Master of Science in Computer Science** | Full Sail University, Winter Park, FL — In Progress

Relevant coursework: Machine Learning, Natural Language Processing, Simulation Modeling, Reinforcement Learning, Software Engineering

**Bachelor of Arts — Computer Science Minor, Digital Media Concentration**

Bridgewater College, Bridgewater, VA — May 2018 | GPA: 3.54

## SELECTED PROJECTS

**Salamander** — Recommendation Systems / Applied NLP — Full Sail University

- Designed a cold-start book recommendation system on a shared sentence-embedding space (all-MiniLM-L6-v2) using a dual-centroid architecture — a feedback-weighted centroid drives recommendations while a second, unweighted centroid audits genre and author diversity
- Built the full stack with React, FastAPI, and Supabase/pgvector, integrating the Open Library API for book metadata
- Evaluated recommendation quality using precision, recall, NDCG, coverage, and novelty, benchmarked against content-based, collaborative, and hybrid recommendation strategies

*Salamander began as a personal library web app built with React, Supabase, and Auth0, where Row Level Security (RLS) policies and Auth0 token-scoped authentication were implemented before rebuilding it into its current form.*

**Raven** — Time Series Forecasting / Financial ML — Full Sail University

- Developed a Mixed-Frequency Transformer to forecast short-term CAD/USD movement from a geopolitical risk (GPR) index, benchmarked against ANN and LSTM baselines
- Tested three input conditions — no GPR signal, GPR resampled to daily frequency, and GPR at its native mixed frequency — to isolate whether preserving the signal's real cadence improves forecast accuracy
- Built the data pipeline in Python with GARCH volatility modeling (arch) and TensorFlow/Keras, cross-verifying CAD/USD data pulled from FRED against Yahoo Finance

**Smart City Simulation** — Reinforcement Learning / Simulation Modeling — Full Sail University

- Engineered a multi-agent urban environment using SimPy on a 65×65 grid, modeling traffic flow, resource allocation, and emergency dispatch with Q-learning (Stable-Baselines3) for autonomous agent optimization
- Achieved quantified improvements in routing efficiency through RL training; exported agent telemetry to Unity 3D via WebSocket pipeline for real-time 3D visualization
- Documented methodology and empirical results in a peer-reviewed-style IEEE-format research paper

**GPT-4o-mini Agentic Chatbot Server** — AI Engineering / Backend — Full Sail University

- Built a production-grade conversational AI system using FastAPI, OpenAI function calling, and async streaming responses with full multi-turn conversation state management
- Implemented long-term memory using ChromaDB vector database and a decorator-based tool registration framework for extensible capability injection
- Delivered rate limiting, content moderation guardrails, and a pytest evaluation harness ensuring correctness across edge cases

**Generative AI Summarization System** — NLP / LLM Evaluation — Full Sail University

- Designed an abstractive text summarization pipeline using Google's Gemma-2-2b-it on the CNN/DailyMail benchmark dataset with HuggingFace Transformers
- Executed systematic ablation study comparing few-shot vs. chain-of-thought prompting across temperature and context length; few-shot yielded the highest ROUGE-L scores across all configurations
- Surfaced non-obvious finding: reduced context window counterintuitively improved ROUGE-L, directly informing production prompt engineering decisions

**Facial Age Prediction with Bias Analysis** — Computer Vision / ML — Full Sail University

- Built a real-time facial age estimation pipeline using DeepFace and OpenCV with threading-based concurrent frame processing for performance optimization
- Identified, characterized, and quantified prediction compression bias — a systematic tendency for the model to regress age estimates toward the population mean — and documented findings in an IEEE-format research paper

**Autonomous Drone Navigation Simulation** — Algorithm Design / AI — Full Sail University

- Implemented a Pygame-based drone simulator with A\* and Dijkstra's pathfinding, dynamic obstacle avoidance, and a finite-state machine for autonomous agent behavior
- Built a real-time HUD displaying agent state, path cost, and navigation decisions for debugging and demonstration

**SELECTED WRITING**

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**Automated Data Collection in Human-Computer Interaction** — IEEE-format paper — Full Sail University — Aug. 2026

- Surveyed automated data-collection methods in HCI research — web server/proxy logs, activity-logging software, instrumented and purpose-built research software, and hybrid log-plus-observation approaches — and evaluated software tools for standardizing interaction logs and comparing automated against human-augmented usability evaluation
- Assessed privacy risk in commercial session-replay tracking, drawing on a survey finding that fewer than one in eight sites disclosed both the tool used and its purpose

**Usability Testing of Different Research Methods in Human-Computer Interaction** — IEEE-format paper — Full Sail University — Aug. 2026

- Compared expert-based, automated, and user-based usability testing methods — including formative vs. summative and remote vs. in-person study design — against the field's "five users is enough" sample-size heuristic
- Examined how resulting usability data is visualized, from interaction-effort scoring to eye-tracking heatmaps, and traced AI's growing role on both the testing and visualization side of the field

**TECHNICAL SKILLS**

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|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Languages        | Python, Java, JavaScript, C, HTML/CSS, SQL                                                                                                                          |
| Frameworks       | FastAPI, React, TensorFlow/Keras, Sentence-Transformers, SimPy, Pygame, Unity 3D, Stable-Baselines3, HuggingFace Transformers, OpenAI API, OpenCV, DeepFace, YOLOv8 |
| AI / ML          | Machine Learning, NLP, Recommendation Systems, Time Series Forecasting, Reinforcement Learning, Generative AI, Computer Vision, Prompt Engineering                  |
| Modeling & Stats | Decision Tree, Random Forest, SVR, Logistic Regression, Naive Bayes, KNN, Clustering (k-means, hierarchical), PCA, GARCH (arch), A*/Dijkstra's pathfinding          |
| Evaluation       | ROUGE, BERTScore, NDCG, AUC/ROC, RMSE, R <sup>2</sup> , Precision/Recall, F1, Cohen's Kappa, Confusion Matrix, A/B Testing                                          |
| Data & Auth      | ChromaDB, Supabase/pgvector, Auth0, Row Level Security (RLS) policies, Pandas, NumPy, Jupyter Notebooks                                                             |
| Tools            | Git & GitHub, CLI/Bash, pytest, Adobe Creative Suite, Microsoft Office Suite                                                                                        |

**EXPERIENCE**

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**Web Design Intern** | *Brim and Bold, Orlando, FL* January 2023 – February 2026

- Delivered tailored web solutions for clients using custom JavaScript, CSS, and accessibility-first UI/UX design principles, from requirements scoping through deployment
- Designed and vectorized hand-drawn graphics using Adobe Illustrator and Photoshop for use in production digital products
- Collaborated cross-functionally to translate client requirements into accessible, performant websites aligned with brand and business goals

**Academic Tutor** | *Bridgewater College, Bridgewater, VA* Fall 2016 – May 2018

- Delivered one-on-one and group instruction in core STEM and arts subjects, developing custom study materials tailored to individual learning gaps
- Evaluated student work and provided targeted, constructive feedback to accelerate comprehension and academic performance

**CERTIFICATIONS & PROFESSIONAL DEVELOPMENT**

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**Google IT Support Professional Certificate** — Coursera | June 2022

**Full Stack Engineer Path • Computer Science Path** — Codecademy | 2024